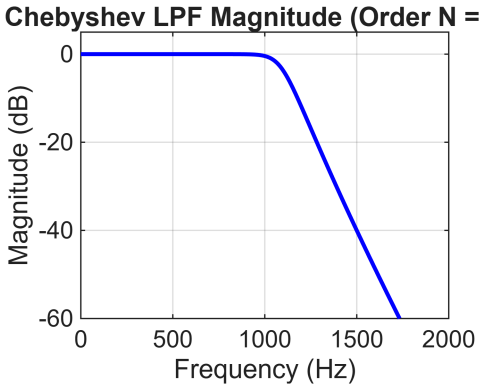
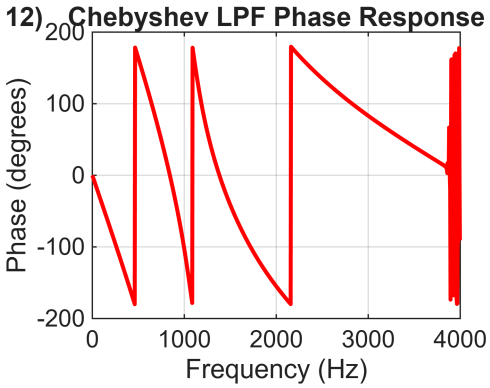
```
subplot(2,2,1);
plot(f, 20*log10(max(abs(H_lp), 1e-10)), ...
    'b', 'LineWidth', 1.5);
grid on;
title(['Chebyshev LPF Magnitude (Order N = ', num2str(N_lp), ')']);
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-60 5]);

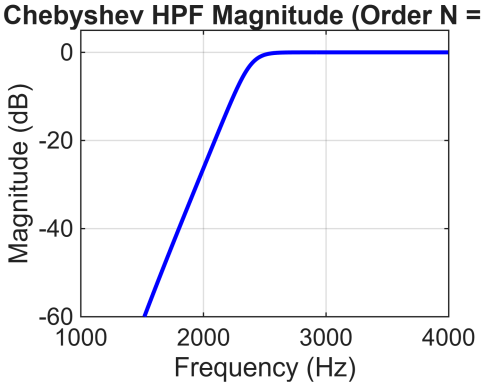
subplot(2,2,2);
plot(f, angle(H_lp)*180/pi, 'r', 'LineWidth', 1.5);
grid on;
title('Chebyshev LPF Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');

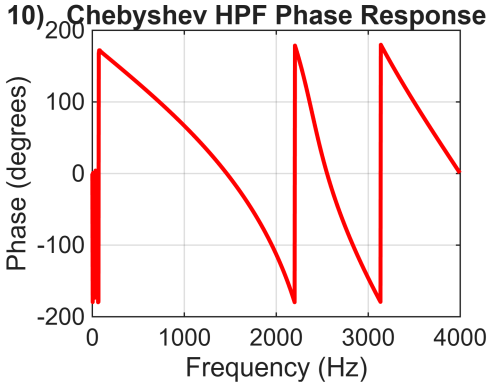
subplot(2,2,3);
plot(f, 20*log10(max(abs(H_hp), 1e-10)), ...
    'b', 'LineWidth', 1.5);
grid on;
title(['Chebyshev HPF Magnitude (Order N = ', num2str(N_hp), ')']);
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-60 5]);

subplot(2,2,4);
plot(f, angle(H_hp)*180/pi, 'r', 'LineWidth', 1.5);
grid on;
title('Chebyshev HPF Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
```


Chebyshev LPF Magnitude (Order N = 12)
Magnitude (dB)
0
-20
-40
-60
0
500
1000
1500
2000
Frequency (Hz)


Chebyshev LPF Phase Response
Phase (degrees)
200
100
0
-100
-200
0
1000
2000
3000
4000
Frequency (Hz)


Chebyshev HPF Magnitude (Order N = 10)
Magnitude (dB)
0
-20
-40
-60
1000
2000
3000
4000
Frequency (Hz)


Chebyshev HPF Phase Response
Phase (degrees)
200
100
0
-100
-200
0
1000
2000
3000
4000
Frequency (Hz)